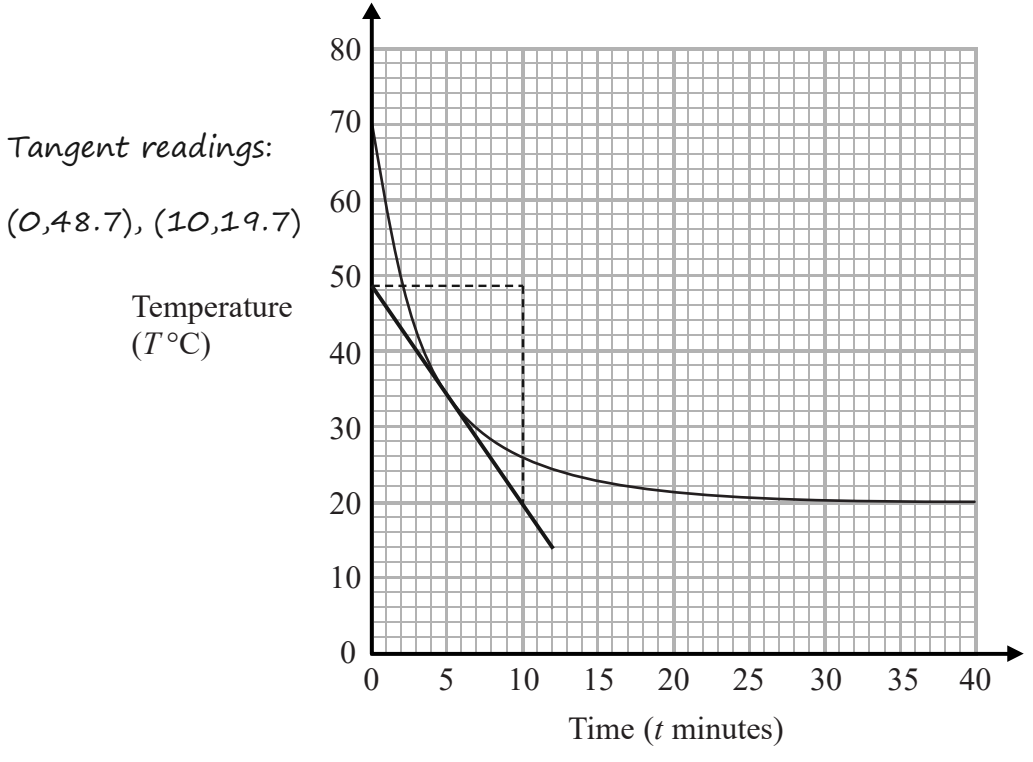
**13** The graph shows the temperature, $T$ °C, of the coffee in a cup at a time $t$ minutes.




**1 mark**
Tangent at t=5
Rise ÷ run

(a) Find an estimate for the gradient of the graph at time 5 minutes.

(19.7 – 48.7) ÷ (10 – 0) ≈ –2.9°C/min.

**1 mark**
About –2.9°C/min

**(2)**

(b) Explain what this gradient represents.

At5 minutes,coffee is cooling at about2.9°C per minute.

**1 mark**
Rate of temperature change at5minutes

**(1)**

**(Total for Question 13 is 3 marks)**